

Pre-inference canvas

(k chosen before inference begins)

Post-inference result

(target `activeCount` $\rightarrow$ [`active`, `Count`], $k^* = 2$)

$k = 1$

```
int <|mask|> = 0;
```

inference

```
int count = 0;
```

$k < k^*$ underflow

1 mask mostly can't fit a 2-token name

$k = 2$

```
int <|mask|><|mask|> = 0;
```

inference

```
int activeCount = 0;
```

$k = k^*$ oracle

canvas size matches target tokens

$k = 4$

```
int <|mask|><|mask|><|mask|><|mask|> = 0;
```

inference

```
int activeUserCountIdx = 0;
```

$k > k^*$ overflow

excess masks $\rightarrow$ filler unnecessary sub-tokens